# Marco Domenico Buttiglione

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## PROFILE

PhD student in Information Technology at Politecnico di Milano, specializing in computer graphics, extended reality (XR), and human-computer interaction. His research targets immersive training systems across different domains, with a particular emphasis on medical applications.

## SKILLS

**PROG. LANGUAGES**
C# • Python • JavaScript
C++ • TypeScript • HTML/CSS

**TECHNOLOGIES**
ReactJS • Unity • TensorFlow
Babylon.js • Vulkan • OpenGL

## LANGUAGES

Italian (native)
English (proficient)

## PROJECTS

**Thoraco VR**
Low-cost VR simulation environment for lung surgery training developed in collaboration with medical professionals.

**Wildcard 2.0**
VR app with eye-tracking for attention training in NDD patients. UAHCI Best Paper Award.

**Paradox!**
2D puzzle platformer with innovative time travel mechanics. Featured on Steam with demo available.

## LINKS

Website:// **Marco D. Buttiglione**
Github:// **MarcoButtiglione**
Steam:// **Paradox!**
YouTube:// **Thoraco VR**

## EDUCATION

### POLITECNICO DI MILANO
PhD in Information Technology
Sep 2024 – Present | Milan, Italy

- Specialized curriculum focused on computer graphics and VR/AR technologies. Advisor: Prof. Marco Gribaudo.

### POLITECNICO DI MILANO
MSc in Computer Science and Engineering
Oct 2021 – Apr 2024 | Milan, Italy

- Thesis: *Soft-Body Simulation for Virtual Reality-Based Thoracoscopic Training*. Supervisor: Prof. Marco Gribaudo. Final grade: *110/110 cum laude*.

### POLITECNICO DI BARI
BSc in Computer Science and Automation Engineering
Oct 2017 – Feb 2021 | Bari, Italy

- Thesis: *Cloud solutions supporting an innovative platform for multimedia content distribution*. Supervisor: Prof. Tommaso Di Noia. Final grade: *110/110 cum laude*.

## EXPERIENCE

### POLITECNICO DI MILANO | Teaching Assistant
Sep 2024 – Present | Milan, Italy

- *Computer Graphics* (Vulkan, Master's degree course in English).
- *Laboratorio di Rappresentazione Digitale* (Bachelor's degree course, 3D CAD modeling for fashion accessories).

### REWIND STUDIOS | Game Developer
Oct 2022 – Present | Milan, Italy

- Directing a heterogeneous indie team of 7 people for the development of *Paradox!*, a 2D puzzle platformer, overseeing game design and mechanics.
- Providing gamification consulting services for external companies, developing interactive applications and video games.

### WIDEVERSE | Frontend Web Developer
Jan 2021 – Sep 2021 | Bari, Italy

- Developed a CRM portal using ReactJS for the civil registry office of the Italian Public Administration.
- Built a 3D web configurator using Babylon.js enabling interactive customization of outdoor blinds.

## PUBLICATIONS

[1] M. D. Buttiglione, F. Guerrera, P. Piazzolla, G. Colombo, E. Ruffini, and M. Gribaudo. Collaborative virtual reality framework for surgical training and simulation. page 661 – 667. European Council for Modelling and Simulation, 2025.

[2] A. Patti, F. Vona, A. Barberio, M. D. Buttiglione, I. Crusco, M. Mores, and F. Garzotto. Training attention skills in individuals with neurodevelopmental disorders using virtual reality and eye-tracking technology. In *International Conference on Human-Computer Interaction*, pages 368–381. Springer, 2024.